```
100 'BAND-PASS WINDOWED-SINC FILTER
110 'This program calculates an 801 point band-pass filter kernel
120 '
130 DIM A[800]                      'A[ ]  workspace for the lower cutoff
140 DIM B[800]                      'B[ ]  workspace for the upper cutoff
150 DIM H[800]                      'H[ ]  holds the final filter kernel
160 '
170 PI = 3.1415926
180 M% = 800                        'Set filter kernel length (801 points)
190 '
200 '                               'Calculate the first low-pass filter kernel via Eq. 16-4,
210 FC = 0.196                      'with a cutoff frequency of 0.196, store in A[ ]
220 FOR I% = 0 TO 800
230   IF (I%-M%/2) = 0 THEN A[I%] = 2*PI*FC
240   IF (I%-M%/2) <> 0  THEN A[I%] = SIN(2*PI*FC * (I%-M%/2)) / (I%-M%/2)
250   A[I%] = A[I%] * (0.42 - 0.5*COS(2*PI*I%/M%) + 0.08*COS(4*PI*I%/M%))
260 NEXT I%
270 '
280 SUM = 0                         'Normalize the first low-pass filter kernel for
290 FOR I% = 0 TO 800               'unity gain at DC
300   SUM = SUM + A[I%]
310 NEXT I%
320 '
330 FOR I% = 0 TO 800
340   A[I%] = A[I%] / SUM
350 NEXT I%
360 '                               'Calculate the second low-pass filter kernel via Eq. 16-4,
370 FC = 0.204                      'with a cutoff frequency of 0.204, store in B[ ]
380 FOR I% = 0 TO 800
390   IF (I%-M%/2) = 0 THEN B[I%] = 2*PI*FC
400   IF (I%-M%/2) <> 0  THEN B[I%] = SIN(2*PI*FC * (I%-M%/2)) / (I%-M%/2)
410   B[I%] = B[I%] * (0.42 - 0.5*COS(2*PI*I%/M%) + 0.08*COS(4*PI*I%/M%))
420 NEXT I%
430 '
440 SUM = 0                         'Normalize the second low-pass filter kernel for
450 FOR I% = 0 TO 800               'unity gain at DC
460   SUM = SUM + B[I%]
470 NEXT I%
480 '
490 FOR I% = 0 TO 800
500   B[I%] = B[I%] / SUM
510 NEXT I%
520 '
530 FOR I% = 0 TO 800               'Change the low-pass filter kernel in B[ ] into a high-pass
540  B[I%] = - B[I%]                'filter kernel using spectral inversion (as in Fig. 14-5)
550 NEXT I%
560 B[400] =  B[400] + 1
570 '
580 '
590 FOR I% = 0 TO 800               'Add the low-pass filter kernel in A[ ], to the high-pass
600   H[I%] = A[I%] + B[I%]         'filter kernel in B[ ], to form a band-reject filter kernel
610 NEXT I%                         'stored in H[ ] (as in Fig. 14-8)
620 '
630 FOR I% = 0 TO 800               'Change the band-reject filter kernel into a band-pass
640   H[I%] = -H[I%]                'filter kernel by using spectral inversion
650 NEXT I%
660 H[400] =  H[400] + 1
670 '                               'The band-pass filter kernel now resides in H[ ]
680 END
```

TABLE 16-2